Yanning DAI

PhD Student | KAUST Centers of Excellence

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Profile

Ph.D. student advised by Prof. Jürgen Schmidhuber, focusing on Reinforcement Learning and Self-Improving Agents. My research spans LLM-based Program Evolution, Robot Learning, and Human Motion Simulation. My publications have received 900+ citations on Google Scholar.

Education

Ph.D. in Computer Science *Jun 2024 – Present*
King Abdullah University of Science and Technology (KAUST), Saudi Arabia GPA: 3.9/4.0
Advisor: Prof. Jürgen Schmidhuber

M.S. in Electrical Engineering *Sep 2020 – Jul 2023*
Beihang University, China GPA: 3.8/4.0
Advisor: Prof. Shuo Gao

B.E. in Electrical Engineering *Sep 2016 – Jul 2020*
Beihang University, China GPA: 3.8/4.0

Selected Publications

1. Efficient Morphology–Control Co-Design via Stackelberg Proximal Policy Optimization.
Yanning Dai*, Yuhui Wang*, Dylan R. Ashley, and Jürgen Schmidhuber.
***ICLR**, 2026.* Paper Project

2. EvoAlloc: A Self-Evolving Resource Allocation Agent for Efficient Program Evolution.
Yanning Dai, Yuhui Wang, Nanbo Li, Wenyi Wang, and Jürgen Schmidhuber.
Preprint, arXiv:2610.12086, 2026. Paper

3. Wearable-Informed Generative Digital Avatars Predict Task-Conditioned Post-Stroke Locomotion.
Yanning Dai, Chenyu Tang, Ruizhi Zhang, Wenyu Yang, Yilan Zhang, Yuhui Wang, et al.
Preprint, arXiv:2512.14329v2, 2026. Under review at Science Advances. Paper

4. A Unified Framework for Rethinking Policy Divergence Measures in GRPO.
Qingyuan Wu, Yuhui Wang, Simon Sinong Zhan, **Yanning Dai**, Shilong Deng, Sarra Habchi, et al.
Preprint, arXiv:2602.05494, 2026. Paper

5. OASIS: Online Anchor-calibrated Self-Improvement System.
Wenyi Wang, Firas Laakom, Li Nanbo, **Yanning Dai**, Yuhui Wang, Yifan Xu, Dmitrii Khizbullin, et al.
***Findings of EMNLP**, 2026.* Paper

6. Predictive Semantic Safety: From Visual Physical Reasoning to Safety-Critical Control.
Taekyung Kim*, Salem Fradi*, **Yanning Dai**, Mateusz Ostaszewski, and Jürgen Schmidhuber.
Preprint, arXiv:2609.34356, 2026. Paper Project

7. Advanced Electronics and Artificial Intelligence: Must-have Technologies toward Human Body Digital Twins.
Yanning Dai, Jiaqi Wang, and Shuo Gao.
***Advanced Intelligent Systems**, 2022.* Paper

8. A Flexible Multi-Functional Smart Skin for Force, Touch Position, Proximity, and Humidity Sensing for Humanoid Robots.
Yanning Dai and Shuo Gao.
***IEEE Sensors Journal**, 2021.* Paper

Selected Research Projects

Reinforcement Learning and Robot Control

- **Stackelberg PPO:** Developed a game-theoretic RL algorithm for morphology–control co-design, accounting for how control policies adapt to morphology changes to improve optimization efficiency.
- **Humanoid Motion Learning:** Combined reinforcement and imitation learning to acquire motion styles from limited demonstrations and transfer them to unseen tasks and terrains in simulation.
- **Cross-Embodiment Control (Ongoing):** Developing a benchmark to evaluate whether a single agent can control diverse robot bodies and generalize to morphology changes, measuring both control performance and interaction costs.

LLM Agents for Robotics and Self-Improvement

- **Robot Design and Control with LLM Agents (Ongoing):** Developing tool-using LLM agents that iteratively design LEGO robot structures and control programs using simulation feedback.
- **EvoAlloc:** Developed a self-improving resource allocation agent for LLM-based program evolution, achieving baseline performance with 60% fewer LLM tokens.

Research Experience

Research Intern, *Tsinghua University* *Jan 2024 – May 2024*
Human Digital Twin Simulation and World Models

- Developed a data–physics hybrid framework integrating wearable motion data with physics-based humanoid simulation. Curated IMU recordings and trained patient-specific agents through reinforcement and imitation learning.
- Continuously personalized RL agents using new patient motion data to predict locomotion across unseen tasks and environments through physics-based simulation.

Skills

Programming & Tools: Python, C/C++, Bash, MATLAB; Linux, Git, Slurm
ML & LLM Inference: PyTorch, Hugging Face Transformers, TensorFlow; vLLM, SGLang
Simulation & Robotics: MuJoCo, Gazebo, Unity3D, COMSOL Multiphysics, Universal Robots
Languages: English (Fluent), Mandarin (Native)

Selected Presentations

1. Starfish Robot—One Policy Across Radical Morphologies.
Research presentation, KAUST RISC Lab *Feb 2026*

2. Reinforcement Learning Enabled Personalized Motor Disease Rehabilitation Program Design.
Invited speaker, KAUST Rising Stars in AI Symposium *Feb 2024*

Honors

1. Rising Stars in AI, KAUST AI Initiative (30 Awardees Worldwide) *Apr 2024*
2. National Scholarship, Ministry of Education of China (Top 5%) *Dec 2022*
3. Technological Invention Award, Third Prize, China Instrument and Control Society *Oct 2022*
4. First Prize, China Instrument and Control Society Scholarship (Top 1%) *Nov 2019*

Academic Service

Reviewer: NeurIPS, ICLR, EMNLP, ICRA; IEEE Sensors J., IEEE TNSRE.